Lab experiment-24

Experiment 24. Monkey Banana Problem

Aim

To implement the **Monkey Banana Problem using Prolog**.

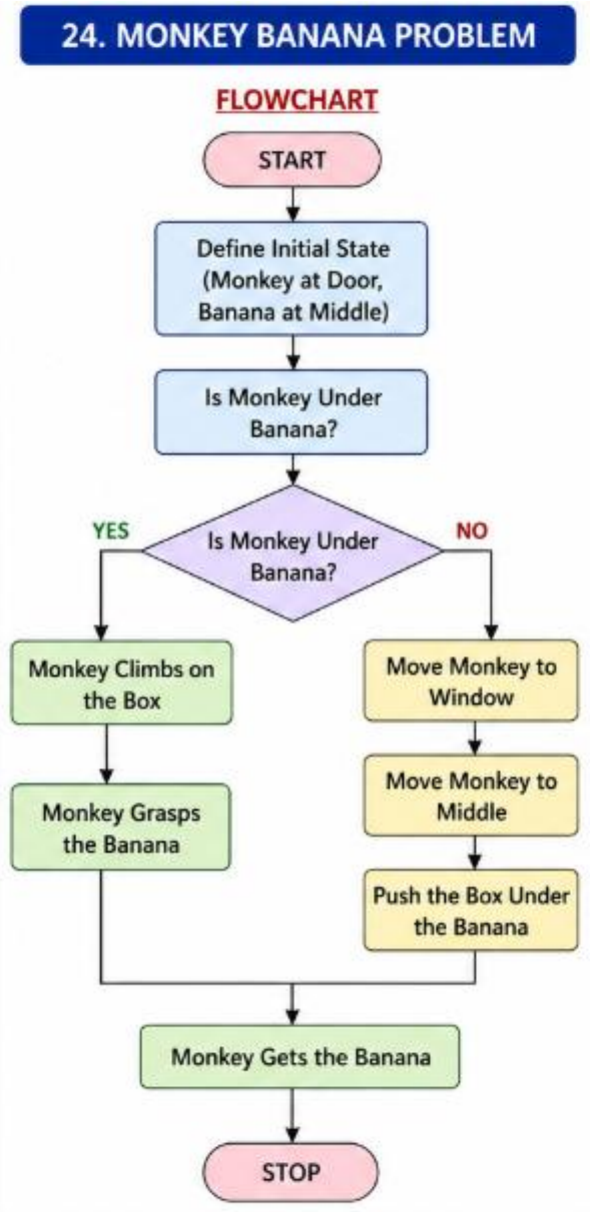
Objectives

- To understand state-space representation.
- To solve a simple planning problem using Prolog.
- To determine the sequence of actions required to get the banana.

Algorithm

1. Define the initial state of the monkey.
2. Define the position of the banana.
3. Check whether the monkey is under the banana.
4. If not, move the monkey.
5. Push the box under the banana.
6. Climb the box.
7. Grasp the banana.

Flowchart



Code

move(monkey, door,

window).

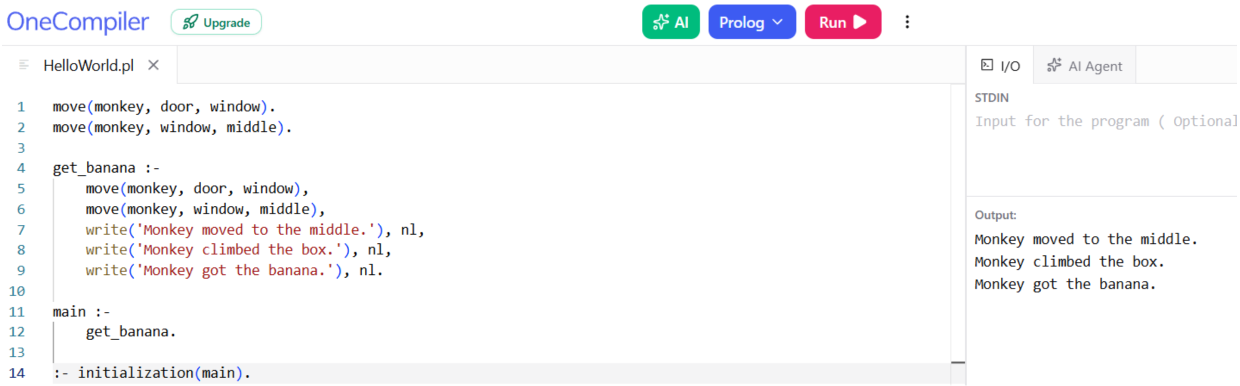
```
move(monkey, window,
middle).

get_banana :-
    move(monkey, door,
window),
    move(monkey, window,
middle),
    write('Monkey moved to
the middle.'), nl,
    write('Monkey climbed
the box.'), nl,
    write('Monkey got the
banana.'), nl.

main :-
    get_banana.

:- initialization(main).
```

Output



d

Result

Monkey moved to the middle.

Monkey climbed the box.

Monkey got the banana.

Conclusion

Thus, the Monkey Banana Problem was successfully implemented using Prolog.